



Leveraging Workarounds for a Problem-Focused Improvement of Business Processes

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Abstract Business process improvement remains a central activity in organizations seeking to address process deviations and enhance operational performance. This paper introduces an innovative and systematic process improvement approach that leverages knowledge derived from recurrent workarounds performed by process participants. Building on prior evidence that such workarounds often originate from perceived goal and process misalignments, a structured procedure is proposed that detects, analyzes, and interprets workarounds to uncover their underlying root causes. The main artifact developed in this study is the Workaround-Driven Improvement Cycle (WDIC), an iterative improvement cycle that connects recurring workaround patterns to targeted improvement directions. To operate this cycle, a complementary improvement decision framework is introduced that links misalignment categories to appropriate improvement directions. The approach was developed using a Design Science Research (DSR) methodology and informed by goal-based analysis, the workaround motivational model, and established categorizations of misfits and process redesign patterns. The WDIC and its integrated decision framework were

developed across three organizational settings, each involving numerous recurring workarounds, with a fourth case providing a full demonstration of the WDIC's practical application. The results demonstrate that systematically investigating workarounds provides focused insights into socio-technical misalignments and supports the design of precise, context-aware process improvements.

Keywords Business process improvements · Workarounds analysis · Goal misalignment · Case study

1 Introduction

Organizations have managed and improved their business processes in various ways over the years. Business Process Management (BPM) employs methods, techniques, and tools to support the design, execution, and control of business processes (van der Aalst 2013). It helps organizations operate effectively and efficiently through continuous discovery, execution, analysis, and redesign of business processes (Dumas et al. 2018), in a cycle of ongoing improvement, seeking to increase organizational stability, productivity, adaptation to competitive markets, and more (vom Brocke et al. 2021; Grisold et al. 2019). The analysis involved in process improvement is considered complex, requiring familiarity with the process under consideration and the organizational setting, as well as vast knowledge and expertise in the field of business processes (Grisold et al. 2021).

One of the symptoms that may indicate an existing problem or misfit that could be solved as part of process improvements is the use of workarounds in the process, where process participants deviate from the process, using ad hoc ways to achieve their goals (Outmazgin et al. 2020).

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Such deviations can be in any aspect of the process, such as activities and their partial ordering rules, use of data, use of technology, and more (Alter 2014; Outmazgin and Soffer 2016). Business process workarounds can be regarded positively as a creative improvisation and a source of innovation (Zamani and Pouloudi 2021; Bartelheimer et al. 2023) offering an accessible solution to the problem at hand. Typically, this positive view may imply that the process can be improved by adopting the practiced workaround, as has been suggested by many researchers (Alter 2015a; Friedman et al. 2014; Beerepoot et al. 2019a).

However, in many cases, workarounds may also impose risks, including reduced quality of products, financial losses, violation of regulations, and potential lawsuits (Röder et al. 2014a). While workarounds may be effective locally for overcoming certain misfits, they may, due to their local and ad-hoc nature, cause problems elsewhere (Röder et al. 2014b; Barata et al. 2015; Beerepoot and van de Weerd 2018). As such, they may not be the best solution to whatever misfit they stem from.

Our premise is that workarounds are performed for a reason and are driven by underlying problems and limitations in the process or in the information system (IS), which together form a socio-technical system comprising technical components, people, organizational processes, and their mutual interactions (Laudon and Laudon 2021). The analysis and understanding of workarounds and the motivation that drives them may hence reveal their root causes. This may form a source of innovation for process improvement, where the workaround and the knowledge gained from it can be considered when examining potential solutions. The workaround may be one of the candidates for such a solution, but not the only one and potentially not the best one.

This direction has already been suggested by several researchers. Outmazgin et al. (2020) suggested a goal-based analysis approach that systematically analyzed workarounds to identify goal conflicts and misalignments associated with them, using the well-established *i** model (Horkoff and Yu 2013). They showed that this analysis can form a basis for business process improvements. Soffer et al. (2023) proposed the theoretical workaround motivational model, which is based on the Theory of Planned Behavior (TPB) (Ajzen 1991). Their findings indicated that the motivation for workarounds arises from the perception of misalignment between organizational, local-unit, or personal goals and the manner in which these goals are realized within business processes. Moreover, workarounds are enabled by a supportive social atmosphere as well as technical and organizational opportunities for workarounds. Therefore, analyzing the root-causes of workarounds as a basis for identifying process or IS improvement opportunities is not new. However, the above

studies focused on the identification of improvement opportunities, while the actual development of an appropriate improvement was not systematically supported.

The aim of this paper is to propose an ongoing improvement process, a *workaround driven improvement cycle* (WDIC), which leverages knowledge about process workarounds and their underlying root causes to enable structured and actionable process improvements. We focus specifically on recurrent workarounds, those that occur repeatedly under similar circumstances and therefore indicate persistent, structurally rooted misalignments. According to Dumas et al. (2018), in the last phase of the BPM lifecycle, “deviations with respect to the intended behavior are identified and corrective actions are undertaken. New issues may then arise, in the same or in other processes, which requires the cycle to be repeated on a continuous basis” (p 24). The WDIC will support the analysis of deviations between intended behavior and actual behavior and the corresponding development of improvement candidates.

To this end, this study adopts the Design Science Research (DSR) approach (Peffer et al. 2007) to develop a structured and systematic procedure for improving business processes, grounded in the detection and analysis of process workarounds and their underlying root causes. The main artifact of this study is the WDIC. To enable its practical implementation, the study also introduces a secondary, but essential, artifact, which is an *improvement decision framework* that supports the selection of appropriate solutions for a given identified workaround.

The framework was developed based on the motivational model of Soffer et al. (2023), according to which workarounds arise mainly from goal and process misalignments. Although some studies point to additional drivers of workarounds, such as technical mishaps or temporary failures (e.g., Alter 2014; Zimmermann and Rentrop 2014), our focus here is on workarounds that are systematically performed, arising from perceived goal and process misalignments. Three case studies were conducted to identify such misalignments and propose appropriate improvement solutions. The identified misalignments and improvements were categorized into high-level categories informed by literature on misfit (Strong and Volkoff 2010; van Beijsterveld and van Groenendaal 2016) and process improvement patterns (Mansar and Reijers 2007; Van Looy and Shafagatova 2016; Gross et al. 2021). The resulting framework was evaluated and validated through interviews with seven process improvement experts. The complete WDIC, including the integrated decision framework, was demonstrated and evaluated in real-world organizational contexts. By addressing workaround-related misalignments and structured improvement selection, this study offers a practical and theoretically grounded solution. Proposed

improvements were presented to stakeholders and revisited after one year to assess their realized impact on processes.

The rest of the paper is organized as follows. The following section reviews literature as the background of our study. Then, the DSR method is described, followed by the development of the improvement decision framework and its evaluation by Business Process Improvement (BPI) experts and organizational stakeholders. Next, the complete developed WDIC, including the framework, is presented, and its use is explained in detail, demonstrated and validated through an additional case study, where the WDIC was evaluated within an organizational context. Finally, conclusions are drawn.

2 Theoretical Background

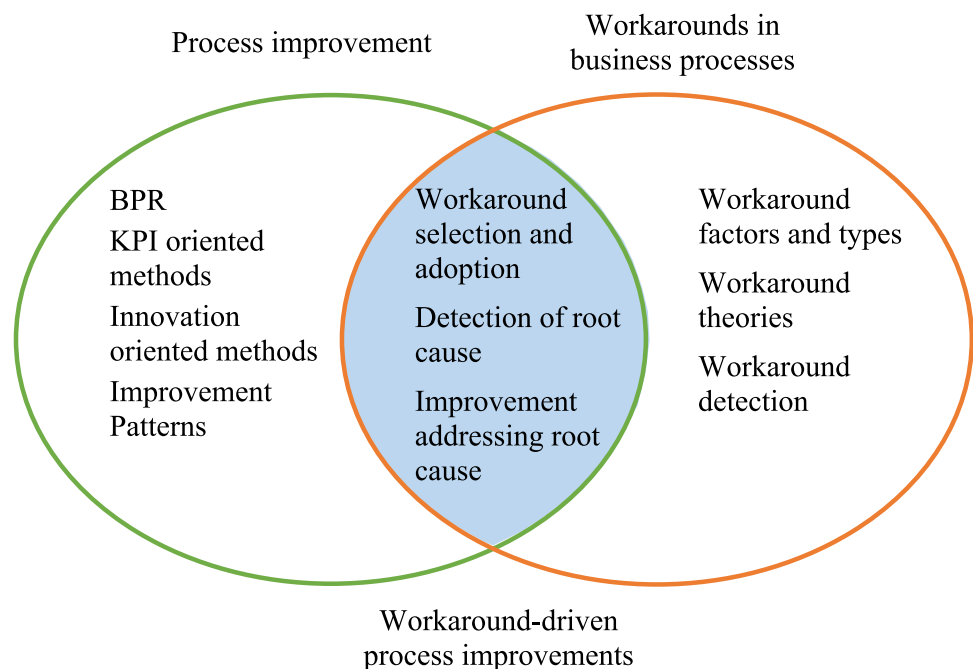
To provide a clear and accurate overview of the theoretical background for this study, we review two primary research domains. The first one is of process improvement methods and techniques, while the second one examines methods and techniques used to detect and investigate workarounds performed by process participants. We then concentrate on the intersection of these domains, specifically how knowledge of workarounds can be utilized to improve processes. This includes determining whether to adopt workarounds as part of the official process, and if not, understanding how their investigation can contribute to process improvements. Figure 1 illustrates the main areas of the theoretical background.

2.1 Process Improvement

Over the years, numerous methods and approaches have been developed for BPI. These approaches have gradually evolved into a comprehensive body of work within the realm of BPM (van der Aalst 2013), including methods, techniques, and conceptual frameworks for the design and implementation of process improvements (Vanwersch et al. 2016; Martins and Zacarias 2017; Satyal et al. 2019; Grisold et al. 2022). Furthermore, case study-based research was conducted in specific domains, such as healthcare and retail (Seethamraju and Marjanovic 2009; Beerepoot and van de Weerd 2018), and systematic literature reviews (SLR) were conducted, aiming to develop a knowledge base that may help organizations improve their business processes (Kettler et al. 2019; Schmitt and Hörner 2021). Well-established approaches to ongoing process improvements exist, including Six Sigma for quality and variation reduction (Harry and Schroeder 2006), Lean Management for waste elimination (Womack and Jones 1997), Total Quality Management (TQM) for organization-wide quality culture (Deming 2018), Total Productive Maintenance (TPM) for equipment (Nakajima 1988), and Business Process Reengineering (BPR) for radical process redesign (Hammer and Champy 2009). These approaches offer methods that serve as valuable guides for system analysts during process improvement.

The complexity of process improvement and its knowledge-intensive nature have long been acknowledged. For instance, Seethamraju and Marjanovic (2009) describe process improvement as a set of coordinated, collaborative,

Fig. 1 Theoretical background illustration



and contextualized knowledge management processes. This implies a need for knowledge co-creation, as opposed to the emphasis on the design of a new process model, taken by lifecycle models (de Morais et al. 2014). Specifically, better process models do not automatically yield better processes (Dumas 2015), hence the focus in BPI should be on improving business processes themselves, rather than improving their models. This reflects the common gap between the formal model and its real-life execution, which is often shaped by human behavior and organizational context. Our WDIC aims to leverage workaround knowledge of real-life executions as part of a process lifecycle.

A means of evaluating process performance and its improvement is to measure its Key Performance Indicators (KPIs), related to typical process performance dimensions such as time, quality, costs, and flexibility. However, not all processes have clear KPIs and data for precise measurement (Dumas et al. 2018). Some process KPIs can be measured quite easily, such as cycle time and cost, which may be measured according to a predefined service level agreement (SLA). The quantification of other KPIs may be more difficult and time-consuming; for example, customer satisfaction may require collection of customer experience survey data, product evaluations, loyalty analyses, etc. (van der Aalst et al. 2016). It follows that BPI cannot rely solely on KPIs and performance measures, and additional indicators are needed for identifying and designing improvement solutions.

An additional research stream focuses on BPI patterns that provide practical guidance for identifying and structuring improvement actions. Mansar and Reijers (2007) identified widely used best practices for BPR and demonstrated their impact on performance outcomes. Van Looy and Shafagatova (2016) structured performance measurement indicators to support process evaluation, while Gross et al. (2021) introduced the business process design space to systematically explore alternative process designs along multiple dimensions. Together, these contributions provide a foundation for categorizing improvement opportunities and supporting structured decision-making in process improvement initiatives.

2.2 Workarounds in Business Processes

The general phenomenon of workarounds has been studied by many researchers and in many domains, such as healthcare, manufacturing, technological and public administration (Fraczkowski et al. 2021; Larsson et al. 2021; Röder et al. 2014; Campbell 2012) illustrating the broad relevance and contextual diversity of workaround phenomena across sectors. Many issues have been indicated as engendering motivations that lead to workarounds,

such as technological malfunctions and problems (Zimmermann and Rentrop 2014; Spierings et al. 2017), business processes that seem too slow or misaligned with IS (Bagayogo et al. 2013), missing required information on systems or processes (Berente and Yoo 2012), obstacles to doing work in a preferred manner (Campbell 2012), misalignments or conflicts of personal goals with organizational ones (Outmazgin et al. 2020), strict regulations and standards (Röder et al. 2014b), situational constraints, or anomalies that render the performance of work activities difficult (Alter 2014). Nonetheless, not all workarounds necessarily indicate that a process is malfunctioning or misaligned. Some are ad hoc technical fixes addressing transient issues (e.g., system outages), which do not reflect persistent process deficiencies (Zaheri et al. 2025). In this study, we focus on a specific subset of workarounds that deviate from the formally prescribed business process, as defined by organizational procedures, information systems, or regulatory guidelines, and that occur in a recurrent manner, thus forming a workaround pattern. These distinctions are important because only those workarounds that repeatedly deviate from known practices in pursuit of process goals serve as reliable indicators of structural misalignments (Pinto et al. 2018). In contrast, ad hoc technical fixes, such as one-time responses to system outages, reflect momentary disruptions and typically do not warrant durable process redesign (Hanna et al. 2024). Hence, our analysis centers on workarounds which are more likely to reveal persistent misfits and improvement opportunities in the business process context.

An ongoing debate in the literature concerns whether workarounds, which entail deviations but are nonetheless performed while pursuing the process goal (Wheeler et al. 2012), should be considered as a positive or a negative phenomenon. While technology-oriented workarounds such as quick fixes bear little risks and are often considered positively as creative and innovative solutions to existing problems (Alter 2014), workarounds in business processes, even when promoting process goals, are often considered negatively, as a form of intentional non-compliance. They might lead to a lack of operational efficiency and harm the accomplishment of important organizational goals as a result of focusing exclusively on one particular goal while neglecting others (Outmazgin et al. 2020).

Nevertheless, not all deviations from required processes are necessarily harmful. Certain forms of intentional non-compliance, commonly referred to as *beneficial noncompliance*, can enhance flexibility, adaptability, or performance without compromising broader organizational goals (Alter 2015b). Conversely, *detrimental compliance*, the rigid adherence to dysfunctional or poorly aligned procedures, may inhibit desirable outcomes and reduce responsiveness to practical needs. This perspective emphasizes

the need to evaluate workaround behavior within its organizational context and to consider competing priorities (Davison et al. 2020). Note that workarounds are not limited to formal processes but also address unofficial guidelines or best practices. For example, employees may repurpose attributes in information systems or work around overly restrictive IT usage policies when these prevent timely service delivery. Although such guidelines are not formal mandatory processes, deviating from them still constitutes a workaround, as it diverges from expected organizational behavior (Alter 2015b).

At the same time, it is important to distinguish workarounds from other forms of nonstandard technology use that may appear similar but differ conceptually. Phenomena such as shadow IT, the use of IT systems and solutions without formal organizational approval (Zimmermann and Rentrop 2014) or feral IS, which are unofficial systems developed by end-users (Price and Shanks 2005), share characteristics with workarounds but fall outside the scope of this study. Similarly, the individual use of generative AI tools (e.g., ChatGPT, Gemini) (Wang et al. 2023), low-code/no-code platforms (Davison et al. 2024), robotic process automation (RPA) (Kim 2023), and spreadsheet-based personal solutions (Whelan and Teigland 2013) may support or replace certain tasks, but are only considered workarounds in our context when they constitute intentional and situational deviations from established organizational processes aimed at resolving perceived misalignments or constraints.

Accordingly, we focus on workarounds that emerge within the operational context of established business processes, and which reflect employees' attempts to achieve process goals in ways not prescribed by the known process (Alter 2014; Beerepoot et al. 2019b; Whelan and Teigland 2013). This distinguishes them from highly visible and coordinated adaptations of enterprise systems (e.g., ERP), which may be known to and even supported by management. Such collaborative bypasses reflect local adaptations driven by misfits between global mandates and local realities but are better understood as governance or customization practices rather than "workarounds" in our definition (Malaurent and Karanasios 2020; Davison et al. 2021).

Most studies focus on detecting and learning workaround situations through qualitative techniques such as interviews and observations. Outmazgin (2012) identified situational factors that are typical of workaround situations and six patterns of business process workarounds. Beerepoot and van de Weerd (2018) proposed an approach for improving work systems based on workaround knowledge that was built on structured snapshots in order to capture workaround motivation, impact, possible actions, and the participants involved. A few studies addressed workaround

detection through quantitative techniques such as Process Mining (PM), identifying workarounds through the investigation of process event logs. For example, Outmazgin and Soffer (2016) performed a PM-based investigation of workarounds and found correlations between characteristics of the situation (e.g., activity duration, number of participants) and the frequency of workarounds. Weinzierl et al. (2020) suggested a deep learning-based approach for automatically detecting workarounds in event logs by using a classifier that maps a process instance to a type of workaround. Van der Waal et al. (2025) suggested the SWORD framework, which is a semi-automated detection approach that uses 22 patterns to identify potential workarounds in event logs. PM techniques can be very useful in workaround detection using a quantitative approach, which may also indicate workaround frequency and, as a result, their significance to the organization. However, not all workarounds can be detected by PM techniques (Outmazgin and Soffer 2016), which can provide only indications rather than a full and conclusive detection. In summary, interviews and observations may contribute to the collection of rich data on workarounds but they are very labor-intensive and require access to process participants, whereas PM techniques cannot cover all workarounds. Therefore, mixed methods for the detection and investigation of workaround situations may be considered, as recent studies highlight that both qualitative and quantitative approaches are required to capture the implications of workarounds on process execution and event data (Bartelheimer et al. 2025).

2.3 Workaround-driven Process Improvements

Attempts have been made to suggest how workarounds can serve as a guide for process improvements (McGann and Lyytinen 2008; Röder et al. 2015; Boudreau et al. 2016). For example, Röder et al. (2014a) investigated situational factors of workarounds in three case studies and found that workarounds entail a personal risk–benefit analysis by the employee. If the realized benefits outweigh the situational risks, it might be perceived as a process improvement. However, this perception is from a personal rather than an organizational view. Barata et al. (2015) presented an enterprise-wide framework to deal with risks and workarounds, capturing formal and informal "shadow" practices. They uncovered and confirmed the benefits of integrating workarounds in risk-based thinking to overcome limitations of formally defined processes. Beerepoot et al. (2019b), based on an explorative multiple-case study in five organizations of different types, proposed a set of contextual activities for each stage of a BPI project for recommending process improvements considering observed workarounds. Van de Weerd et al. (2019)

developed a conceptual framework that links workaround features to categories of workarounds and then to certain response actions: prevention, redesign, adoption, or rejection. Bartelheimer et al. (2023) built on theoretical knowledge on business processes and organizational routines to study how workarounds spread within organizations, act as generative mechanisms for bottom-up process innovation and reveal organizational drifts. Their findings suggest that workarounds serve as mechanisms for process change from the bottom-up, complementing top-down process design and intervention efforts. When workarounds do not result in negative consequences, they can be harnessed as a source of innovation to enhance both processes and organizational goals. In summary, these examples show the importance of workarounds for improving business processes.

However, as these studies emphasize the workarounds themselves as a source for process improvement, they neglect understanding the underlying reasons for workarounds, which may contribute to better process improvements. In a limited number of studies, attempts were made to explore this direction. For example, Beerepoot et al. (2020) conducted an analysis of power relations in the context of Health Information Systems (HISs), attempting to understand how power dynamics influence the emergence of workarounds. In their research, they identified 51 workarounds and investigated the underlying power relations associated with them. They categorized these power relations into two main types according to their role in the emergence of workarounds: (1) hierarchical differences among actors and (2) constraints imposed by the system itself. Outmazgin et al. (2020) employed a goal-based analysis approach to discern the motivations driving workarounds. Their research showed that the primary motivation for workarounds often stems from a misalignment between goals and process tasks, so process improvements were proposed to resolve misalignments. Thus, whether workarounds are formalized as official practices or employed as temporary fixes, understanding their motivation may contribute to fine-tuning the focus of process improvements. In line with this growing attention to leveraging deviations for improvement, recent research has proposed structured approaches to assess process deviations systematically. For instance, Grohs et al. (2025) introduced a procedural framework for evaluating the desirability of process deviations, providing step-by-step guidance to distinguish between problematic, neutral, and beneficial deviations. This recent work reflects the increasing emphasis on systematic evaluation of deviations as a foundation for more informed and targeted process improvement initiatives. Similarly, Reineke et al. (2025) initiated a research project to develop a method for systematically evaluating the positive and negative

implications of workarounds, further emphasizing the growing shift from identifying to assessing and learning from workaround phenomena.

2.3.1 Goal Misalignment as a Source of Workarounds

Business process management emphasizes a cross-organizational perception, guided by the aim to standardize and prescribe end-to-end procedures (Hammer 2015). These procedures are usually created and implemented mainly according to organizational goals that are derived from the organizational strategy or the regulation with which it must comply (Dumas et al. 2018). Additionally, different local units may have local goals that may be derived from these root organizational goals or from their role requirements and needs. Haag et al. (2022) explored the motivations that drive employees to implement workarounds to organizational policies. They identified three mechanisms that encourage this: (1) Perceived tension – employees experience tension regarding actions that diverge from policies, especially when these actions are necessary to achieve legitimate organizational goals. (2) Deadlock – employees perceive obstacles that hinder the achievement of goals and seek alternative solutions that may deviate from established policies. (3) Actualization – refers to the tendency of goal-oriented individuals to engage in any actions that may deviate from organizational policy in order to achieve goals. Soffer et al. (2023) developed the workaround motivational model based on TPB (Ajzen 1991), extending it with elements that specifically explain the decision to work around processes. The TPB explains intentions to engage in a behavior through three independent forces: (1) personal attitudes toward behavior, which are related to the individual's beliefs and expectations regarding the outcome of the behavior, (2) subjective norms, which are related to social norms and values that guide one's behavior, and (3) perceived behavioral control, which is related to the perceived level of the individual's ability to control the situation. Figure 2 illustrates the adaptation of the TPB to the context of workarounds.

The motivational model distinguishes between elements that motivate workarounds and those that enable them when a motivation exists. In this paper, we focus on the motivating elements as a basis for process improvement. To understand these elements, it is important to recognize that sometimes the specific local-unit goals may not be fully aligned with global organizational goals (e.g., because of trade-offs among goals that are fulfilled by different local units). Thus, in these situations, employees may face a conflict between pursuing organizational goals and pursuing their local-unit ones. Ideally, the business process should realize both the organizational root goals and the local-unit goals. Still, conflicts may arise between the

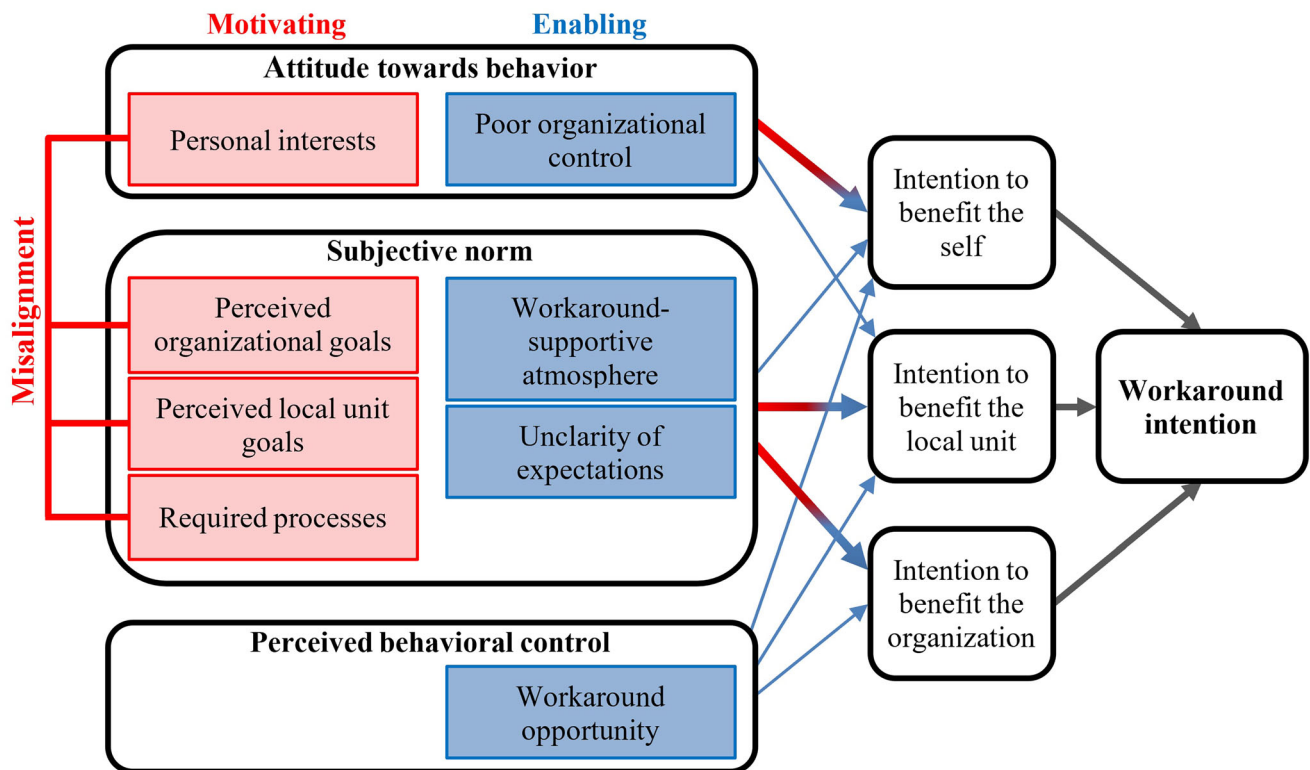


Fig. 2 Workaround motivational model (Soffer et al. 2023)

business process or the IS and the organizational or local-unit goals it is intended to realize (Outmazgin et al. 2020). This aligns with the concept of “misfit” introduced by Strong and Volkoff (2010), who identified six categories of misfit between enterprise systems and organizational needs, such as functionality, data, usability, roles, control, and organizational culture. Van Beijsterveld and van Groenendaal (2016) further developed an analytical framework for diagnosing and resolving such misfits, particularly in ERP implementations. These signify that systemic misalignments between processes, systems, and organizational goals may serve as triggers for workaround behavior.

In fact, misalignment between local goals and the process is inevitable when the goals of the local unit and of the organization are not aligned and the process is designed to realize organizational goals. Considering the local goals, the process is perceived as an obstacle, and this may motivate a workaround (Alter 2014). In the motivational model, the elements comprising the subjective norm (together or separately) can be in misalignment with personal interests (attitudes toward behavior). Any type of misalignment between these elements may motivate a workaround. The enabling elements of the model support performing a workaround when the motivation to do so exists. These include (1) poor organizational control – the individual may consider a workaround as not risky if no

harsh organizational sanctions might be expected, (2) an atmosphere among peers that supports workarounds encourages workaround behavior, (3) unclarity of expectations, which may refer to, for example, contradicting expectations, e.g., make it quickly vs. make it thoroughly, and (4) a workaround opportunity that provides the possibility to work around the process. According to the motivational model, a workaround can be motivated by any type of misalignment between the motivating elements. Once motivation exists, the individual considers the relevant elements that may enable working around the process. As a result, depending on the individual’s perceptions and considerations, the individual may perform a workaround intending to benefit the self, the local unit, or the organization (Soffer et al. 2023).

In summary, workarounds in business processes are often carried out due to perceived obstacles in the process, due to perceptions of misalignments between organizational or local unit goals pursued by the individual. Revealing and investigating the workarounds and their root causes, namely, the perceived misalignments, can create opportunities for tailored and focused process improvements. This direction has not been fully developed so far. Therefore, this study aims to bridge this gap by proposing a way to improve processes based on understanding the underlying root causes of workarounds and the intentions behind them. Considering this, we developed a systematic

workaround-driven process improvement procedure, along with a framework that links possible process improvement directions to goal and process misalignments associated with workarounds. This development was carried out using a DSR approach (Peffer et al. 2007), as described in the following section.

3 Method – Overview

To develop the workaround-based cycle for business process improvement, we took a DSR approach, based on Peffer et al. (2007), following the steps of problem identification, defining solution objectives, design and development, demonstration, and evaluation, as described in Fig. 3.

Our study produced two interdependent artifacts, with the development of one serving as a prerequisite for the other, and each evaluated using different methods. This multi-artifact, multi-evaluation structure also resonates with the echelon-based (eDSR) perspective, recently proposed by Tuunanen et al. (2024), whose steps are consistent with the Peffer et al. (2007) process model. We used the Peffer et al. (2007) process model for describing our DSR process due to its clarity and suitability for our study. When appropriate, we acknowledge eDSR as a means for managing complexity. The theoretical underpinning of our approach was the workaround motivational model (Soffer et al. 2023).

3.1 Problem Identification

Based on an analysis of the literature presented in Sect. 2, we formulated a problem statement:

Workarounds have been identified as a potential source for process improvement. However, while solving problems perceived by employees, they may compromise higher level goals and create risks. As workarounds emerge locally, broader and higher-level alignment is typically not considered.

The problem was, hence, how to leverage workarounds for designing process improvements that will solve existing problems without compromising higher level goals and creating risks.

3.2 Objectives for a Solution

Based on the workaround motivational model (Soffer et al. 2023), workarounds are motivated by perceived goal and process misalignments. Our goal was to develop a systematic process where improvement solutions are devised for mitigating the misalignments that motivate workarounds. This implies three main requirements for the proposed artifact:

Requirement 1: Process improvement solutions will be tailored to address goals and process misalignments that motivate workarounds, consistent with prior findings emphasizing that workaround analysis can reveal structural misfits that guide targeted redesign (Outmazgin et al. 2020; Pinto et al. 2018).

Requirement 2: The development of improvement solutions will take into account the variety of goals that are involved in the identified misalignments, considering organizational priorities, and minimizing risks. This aligns with studies highlighting the need to balance conflicting organizational and local goals when designing process improvements (Haag et al. 2022; van Beijsterveld and van Groenendaal 2016).

Requirement 3: The developed process improvements will be applicable and feasible to implement in organizations. This echoes prior calls for actionable and context-sensitive process improvement methods (Beerepoot et al. 2019b; Mansar and Reijers 2007).

3.3 Design and Development

We decomposed the development of our composite artifact into two parts (or echelons, as in Tuunanen et al. (2024), as each part had its own development cycles):

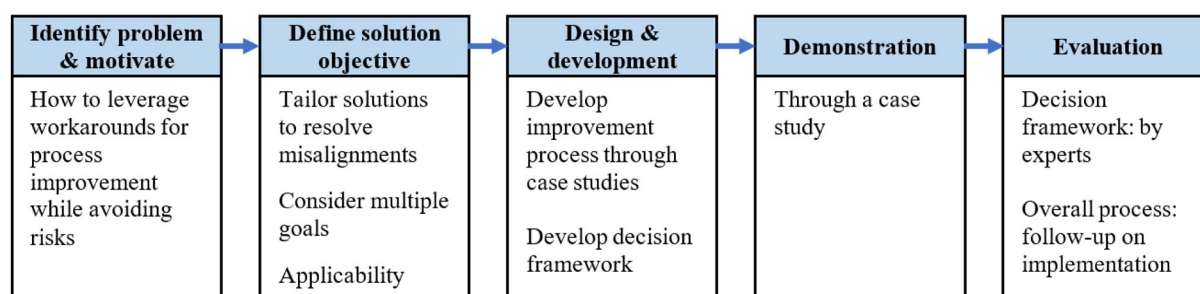


Fig. 3 Research process (following Peffer et al. 2007)

1. Developing an overall systematic process of work-around driven process improvement. This was accomplished through three case studies, where such an improvement process was performed, evaluated with the organizational stakeholders, and refined. This practice-oriented approach was intended for satisfying Requirement 3. In the third case study, the process, as well as its data collection and analysis protocols, has already stabilized to the form of the process we propose in this paper.
2. Developing an improvement decision framework that ties workaround-related misalignments to process improvement directions. This was accomplished by empirically crafting a set of concrete misalignment types that may be associated with workarounds, and concrete improvement directions that can alleviate them. Specifically, the qualitative data collected in the three case studies were analyzed, focusing on identification and analysis of workaround-related misalignments and possible solutions tailored to address them. The resulting sets were composed into a framework that associates misalignment categories with solution directions. Anchoring the framework in workaround-related misalignments ensures that the solutions will address these misalignments (Requirement 1), while considering the variety of relevant goals (Requirement 2). The framework was refined and validated (a) based

on existing categories in the literature, and (b) through interviews with experts (see Fig. 4).

3.4 Demonstration

To demonstrate the overall process, including the decision framework, we conducted a fourth case study, where the proposed process was followed, and improvement solutions were designed using the decision framework.

3.5 Evaluation

The improvement decision framework was evaluated qualitatively by experts, who assessed its usefulness and usability as a tool for process experts.

The WDIC process was evaluated based on the usefulness of its results. This was assessed based on the feedback given by case study stakeholders upon introduction of process improvements and by following up on the actual implementation of the proposed improvements, revisited after a period of one year.

4 Developing an Improvement Decision Framework

As explained in Sect. 3, the aim of the three case studies was twofold. First, by repeatedly performing a process that starts from identifying and analyzing workarounds,

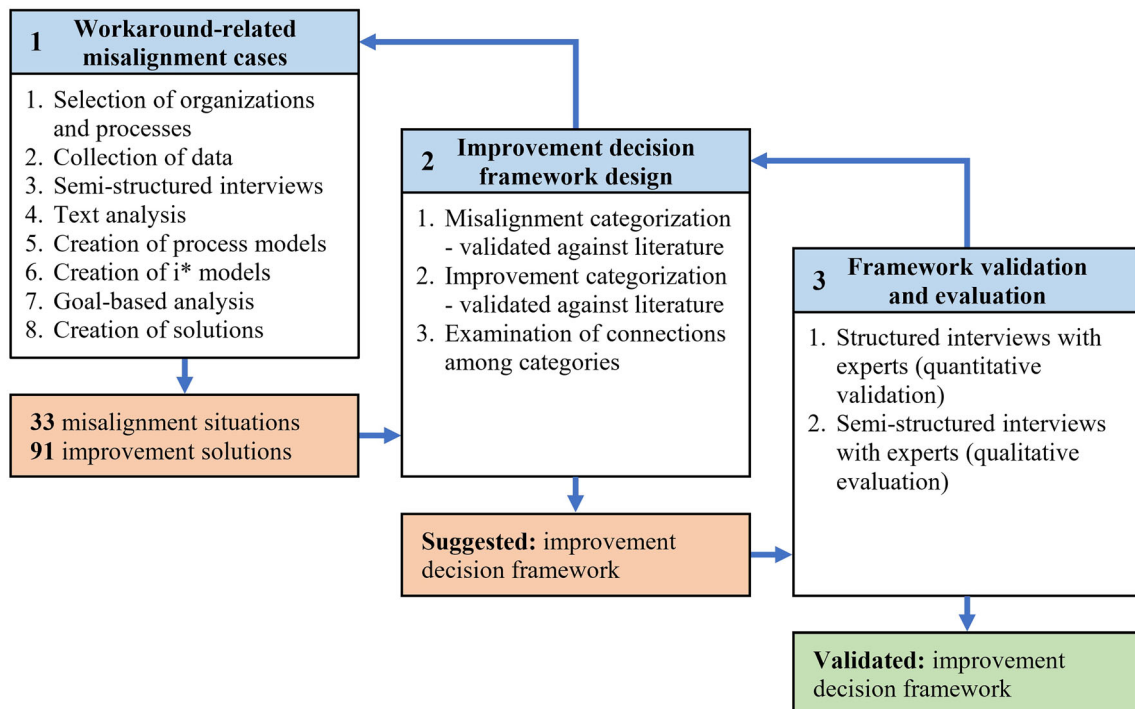


Fig. 4 Improvement decision framework development process

identifying the goal-process misalignments that motivate them, and finally proposing improvements, we aimed to iteratively refine this process as well as its methods and protocols, to become an applicable procedure that can be followed in practice. To this end, we selected our case studies from different organizations and domains. Second, we aimed to use the data collected in these case studies as a basis for developing a decision framework to guide the tailoring of improvement solutions to the identified work-around-related misalignments.

In general, we performed the case studies using the case study research approach (Yin 2013). The first case study builds on the previous work of Outmazgin et al. (2020). In the current study, it was expanded to include a detailed and comprehensive analysis of all revealed misalignment situations, along with systematic proposals for solutions to each situation. Our focus in all studies was on analyzing the organizational environment and the actual processes with the aim of capturing and investigating goal–process misalignments that motivate workarounds in each process and of examining solution options for each uncovered misalignment. First, organizations and processes were selected, and then, the selected processes were comprehensively investigated and documented through document and data collection and analysis, semi-structured interviews that were transcribed and analyzed (Runeson and Höst 2009), and the creation of a Business Process Model and Notation (BPMN) model of all processes (Wohed et al. 2006). Most workarounds were detected through this analysis. Then, to detect additional workarounds and capture their associated misalignments, we used the goal-based analysis approach, which was based on the creation and analysis of *i** models. These models encompass several types of elements, such as process actors, their perceived goals, process tasks, and their relationships. They were instrumental in capturing misalignment situations by assessing two levels of negative contribution (*Hurt or Break*) between them (Outmazgin et al. 2020). Finally, we devised improvement solutions for each captured misalignment situation, supported by the relevant literature.

4.1 Workaround-Related Misalignment Case Studies

Step 1 – Selection of organizations and processes: We selected three organizations of different sizes and business domains, as shown in Table 1, which allowed us to achieve data diversity for the analysis. To gain an understanding of the core functions and processes of each organization, we conducted an initial interview with relevant stakeholders in management positions. Next, we selected three cross-organizational processes to which management gave a high level of attention, where workarounds were known to be

commonly performed, to gain: (1) the cooperation of the process participants and the relevant stakeholders; (2) a high level of data diversity, since these types of processes are considered complex and can involve many interactions between participants and organizational units; and (3) the willingness of the organizations' management to implement or examine the process improvement proposals in the organizational setting for evaluation and validation.

The selected organizations (Table 1) represent diverse sectors and process contexts. Organization 1 is an IT integration company providing technological solutions and services to industrial customers, including data centers, communication systems, cybersecurity, hosting, and internet service solutions. The selected process, *Sales forecast*, is a core process used for promoting sales activities and strategic planning. Organization 2 is a corporation providing feeding solutions and services for various types of farm animals. The selected process, *Customer complaint management*, is critical due to its implications for animal nutrition and the importance of timely and effective response to complaints in this domain. Organization 3 is an academic college offering academic degrees in various fields and is also involved in research and development. The selected process, *Student registration*, is a central process that significantly impacts the institution's budget and strategic decisions.

Step 2 – Collection of data: We collected any type of information that would help us understand the actual processes and the roles of their participants. This included formal documentation of procedures and processes, ISO work procedures and reports, implementation and design documents, and event log data (Online Appendix 1, Table B; the appendices are available via <http://link.springer.com>). We also carefully studied the information system, its user interface, functionality, and database schema, which was accomplished with the help of IS and technical staff.

Step 3 – Semi-structured interviews: For each process, the first author conducted interviews using the interview guide taken from Outmazgin et al. (2020), which focused on collecting workaround and goal information and was hence suitable for this study. The interviews were conducted with relevant process participants, including at least two or three process initiators in operational positions, one or two process supervisors, and one or two stakeholders in management positions. In total, the study included 22 interviewees, whose experience in the organizations ranged from 2 to 23 years, as shown in Online Appendix 1, Table A. The interviews included predefined questions while allowing follow-up questions, free conversation, and additional input from the interviewee (Runeson and Höst 2009), including three main parts. The first was focused on understanding the organizational setting and the manner in

Table 1 Overview of case organizations and processes

Org. no	Type or line of business	Size, distribution of sites	Studied process	Name of IS used	Age of IS (years)
1	IT integration solutions and services	1600 employees, part of corporation with 5300 employees	Sales forecast	Local development	5
2	Agriculture	150 employees	Customer complaint management	Manufacturing ERP	12
3	Academic institute	500 employees, 3000 students	Student registration	Academic ERP	10

which the process serves the organizational needs and goals, with respect to the roles and boundaries of the participants in the process and its activities. The second part was focused on interviewee perceptions of the goals the process should realize, considering the goals of the organization and those of the process participants. This allowed us to reveal and understand misalignment situations in the process. The third part dealt with understanding the workarounds that were performed in the process, how they were performed, their reasons and motivation, their detection by management, and management's response. The duration of each interview was 45–90 min and they were conducted in the offices of the organization (see interview guide in Online Appendix 1, Table C). All the interviews were audio-recorded and transcribed, and in addition, phone conversations and emails were exchanged after the interviews to obtain additional explanations and further validation of the interview transcripts.

Step 4 – Text analysis: Text analysis and interpretation of the transcribed interviews were performed by the first author, mainly to understand the business process, identify its goals as perceived by the interviewees, and gain a deep understanding of the workaround situations and their reported motivation. These analyses were subsequently discussed with the other authors and further validated in the next step (see 4.2). Additionally, and in order to obtain a comprehensive understanding of workaround situations, we combined several different viewpoints of the interviewees of the same situation into one complete situation description. For instance, we found that the process initiator and the approver explained the same workaround differently. The findings from this step were arranged in a table that includes the perceived process goals for each case study separately.

Step 5 – Creation of process models: For each process, we created a BPMN model (Chinosi et al. 2012) that presented the normative as-is process (Online Appendix 2), based on the analyzed interview text and the formal process documentation. In addition, a list of workarounds that had been reported in the interviews and spotted in the logs was compiled. The BPMN model and the workaround list were

validated by the relevant process participants and stakeholders through additional meetings and phone conversations. The validated model and list helped us gain an additional deep understanding of the process, its workarounds (e.g., a gap between the formal process documentation and the workarounds that came up), role boundaries, and activities of participants with respect to their formal goals. Additionally, the BPMN models also contributed to our creation of i* Strategic Rationale (SR) models, as shown in the next step.

Step 6 – Creation of i* models: In this step, we followed the modeling guidelines of Outmazgin et al. (2020) with the aim of ensuring consistency in capturing conflicts between organizational goals, local unit goals, and prescribed processes. For each process, we first created an i* Strategic Dependency (SD) model that provided a high-level representation of the actor interactions in the process to understand the dependencies between the actors and the main organizational goal pursued by each of them. We then zoomed into the i* SR model to examine and focus on misalignment detection as explained in Outmazgin et al. (2020). To achieve this systematically, we used the BPMN models that were created in the previous step, and the tables of the perceived process goals revealed through the text analysis. Table 2 provides a brief overview of the i* SR models for each process (the full i* SR models of the processes are shown in Online Appendix 3).

As can be seen in Table 2, the contribution links are of two direction types: positive in two levels (*Help and Make*) and negative in two levels (*Hurt and Break*). These are marked in the table as P for positive and N for negative.

Step 7 – Goal-based analysis: To detect and analyze goal misalignments in the i* SR goals process models, we used the goal-based analysis (Outmazgin et al. 2020), focusing on identifying negative contributions (i.e., *Hurt or Break*) between goals and tasks. As described in Outmazgin et al. (2020), dependency connections among actors create opposing forces that pull actors in conflicting directions. As part of the goal-based analysis using the i* SR models, we identified specific goal misalignments that contributed to workaround behavior in each process.

Table 2 Overview of the processes in i* SR models

i* SR Elements		Count per process		
		Sales forecast	Customer complaint management	Student registration
Actors	Actors/Agents/Roles	17	9	8
	Actor association links	7	5	0
Intentional elements	Goals	14	17	29
	Quality goals	10	10	19
	Tasks	20	19	40
	Resource	1	4	14
	Dependency goals	18	11	22
Social dependencies (between actors)	Dependency quality goals	0	2	1
	Dependency tasks	4	2	1
	Contribution links (Positive/Negative)	27 (P14/N13)	15 (P11/N4)	39 (P23/N16)
Intentional element links	Refinement links	25	28	66
	Qualification links	4	1	2

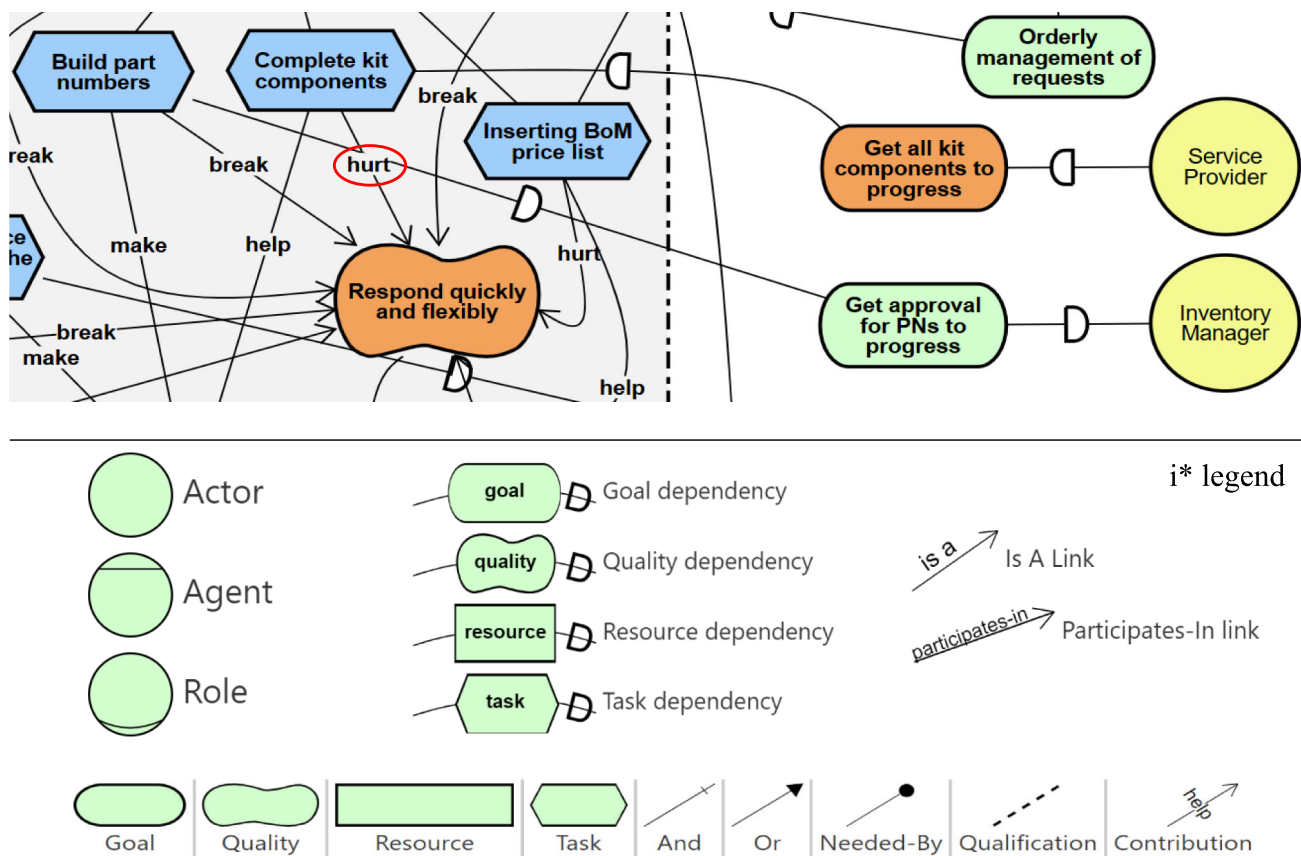
**Fig. 5** Analysis example of a misalignment situation with i* legend

Figure 5, accompanied by the i* legend, illustrates one such misalignment, identified in the *Sales Forecast* process. This example demonstrates how the analytical

method is applied in practice, using the i* modeling approach described above.

As illustrated in Fig. 5, a goal misalignment (in orange) was identified between the goal “*Get all kit components to progress*”, which depends on the task “*Complete kit components*”, and the goal “*Respond quickly and flexibly*”. This misalignment stems from the dependency of the Service Provider actor, who requires the full set of kit components (such as signed formal customer documents) to move forward from quotation to a confirmed sales order. The current process necessitates obtaining full approval of all kit components at an early stage, although typically, not all documents are necessary at this point. This causes delays, impairing the customer manager’s ability to respond promptly. In Fig. 5, this appears as a “*Hurt*” negative contribution to the goal of the customer manager to respond quickly and flexibly to customers. Consequently, the customer manager employs a workaround by entering incomplete or fictitious documents into the IS to advance the quotation, leading to potential blockages in later stages when these documents are fully validated. By applying this analysis to each process, we revealed and analyzed 33 misalignment situations in total – see tables A, B and C in Online Appendix 4.

Step 8 – Creation of solutions: In this step, we proposed specific process improvements for each identified misalignment situation, with the aim of solving or reducing the misalignments that motivated workarounds. The development of improvements was first guided by the investigation of the workarounds and identification of their root causes, namely, the relevant misalignments, as well as the enabling elements. Improvement suggestions for resolving these root causes were initially drawn from the experience of the first author, who has over 18 years of practical experience in this field. This was complemented by an examination of relevant literature to find additional possible solutions (Mansar and Reijers 2007; Dumas et al. 2018; van der Aalst et al. 2016; Kettler et al. 2019). The observed workarounds were also considered among the possible solutions (although in most cases, they turned out to be too local to solve the actual misalignment, considering all the involved goals). In sum, solution development relied on practitioner experience as the primary foundation, supported by theoretical literature and empirical insights drawn from the investigated workaround. This layered combination ensured both practical relevance and conceptual rigor in the proposed solutions.

Then, we discussed the proposed solutions with the process stakeholders in a series of meetings and phone calls in which the two parties engaged in collaborative thinking about the solutions and considered the relevant implementation details in the organizational setting. For example, in the Sales Forecast process, we proposed a reduction in the strictness of the approval rounds in order to reduce the misalignment between this lengthy procedure and

local-unit goals that require a quick response. The stakeholders translated this proposal to the relevant implementation details in the organization, which were manifested in procedural and technological changes that were approved by the management. We then fine-tuned our proposed solutions and arranged them in a table according to the misalignment situations they resolved. In total, we proposed 91 improvements for the 33 misalignments; that is, in most situations, we proposed more than one improvement.

4.2 Improvement Decision Framework Design

In this phase, to design an improvement decision framework that can provide a solution for the identified problems, we separately categorized the 33 misalignment situations and the 91 improvements that were proposed for addressing them. The categorization involved evolving open and axial coding for each group until saturation was achieved (Flick 2018; Williams and Moser 2019; Miles et al. 2014). As a result, the 33 captured misalignment situations were categorized into 3 main categories, and the 91 proposed improvements into 10 main categories. Further, in the process of selective coding, we hypothesized associations (that were later statistically examined) between the misalignment and the improvement categories.

Step 1 – Misalignment categorization: To categorize the misalignment situations, we learned the root underlying reasons of the negative contributions (*Hurt or Break*) in the 33 conflicts and misalignment situations (Online Appendix 4). This helped us gain the comprehensive understanding required to generalize the specific misalignments caused by specific issues, to higher-level abstract categories. Then, we grouped misalignment situations based on similarity of meaning and direction, until three main categories, that could not be further unified, were obtained. Saturation was obtained when all the misalignment situations were classified into the three predefined categories, with no need for additional categories or refinement of existing ones (Miles et al. 2014). Then each category was examined and mapped to the six categories of misfit described in Strong and Volkoff (2010) and the misfit analytical framework described in van Beijsterveld and van Groenendaal (2016), both of which propose types of misfit that closely resemble the misalignment categories revealed in this study, thereby supporting the validity of the categorization performed. These works offer a theoretical basis for the misalignment classification. We illustrate the categories using the following examples.

Regulation category example (Situation 1.11, Table A, Online Appendix 4, and also Fig. 5): The identified misalignment stems from a formal procedure, designed to mitigate risks associated with undesirable customer

engagements. It was hence classified under the regulation/procedural category.

Technological category example (Situation 1.13, Table A, Online Appendix 4): To achieve the goal “Respond quickly and flexibly,” the customer managers are required to invite a presale engineer through the IS to create a Bill of Materials (BOM) in the quotations that should be sent to the customers. In this case, the misalignment stems from the partial support of the IS functionality for inviting presale engineers, as there is no appropriate prioritization mechanism in the system. In fact, inviting a presale engineer as required may lead to the opposite result and delay the process. Therefore, this misalignment was classified under the *technological/implemental* category. To clarify the scope of this category, we note that the analysis is focused on goal/process misalignments, some of which stem from technological limitations. Namely, a misalignment that is identified between using the system (compliantly) and some goal of the process, which is attributed to a shortcoming in the system. However, situations requiring a simple system repair, such as errors or technological malfunctions, are not included in this category.

KPIs category example (Situation 1.1, Table A, Online Appendix 4): The customer manager’s goal “Achieve high sales targets performance” was found to be in misalignment with the management goal “Obtain a real picture of the expected sales for planning.” The future sales figures available to management are based on reported sales forecast (proposals entered into the system) and depend on the customer managers’ reports. However, the performance assessment and rewards of customer managers are based on closing deals only and would not benefit from the proposals being reported before the deals are closed. Since this misalignment stems from inconsistencies in performance assessment measures between one or more actors, it was classified in the *KPIs* category.

A summary of the three main categories, along with their corresponding mappings to the relevant literature, is presented in Table 3.

Step 2 – Improvement categorization: To establish categorization for the proposed solutions, each specific improvement was initially evaluated based on the BPR framework components introduced by Mansar and Reijers (2007), the performance perspectives introduced by Van Looy and Shafagatova (2016), and the BPR dimensions introduced by Gross et al. (2021). Our primary attention was directed toward internal organizational improvements. Consequently, components related to customers and the external environment were regarded as of less significance within the scope of our study. Given our defined scope, the components described in Mansar and Reijers (2007) and the performance perspectives described in Van Looy and

Shafagatova (2016) have been fully mapped to most of our proposed improvements, while the dimensions described in Gross et al. (2021) were found to partially overlap the basic components, albeit with supplementary insights. To examine additional improvement directions, we used additional sources (Kettler et al. 2019; Schmitt and Hörner 2021; Dumas et al. 2018). This process helped us create several main improvement categories and classify specific improvements into known and formal categories that have already been discussed in the literature. For example, the specific suggested improvements “Introducing threshold conditions for the amount and the profitability rate that require approval” for the sales forecast process and “Setting a level of monetary compensation amount that requires approval” for the customer complaint management process were classified in the category related to process gateways. We continued this process, classifying improvements and refining the categories while considering relevant literature, until saturation was achieved (Miles et al. 2014). That is, all proposed improvements were classified into already existing categories, with no need for additional categories or further refinements. As a result, 10 improvement categories were formed. Table 4 summarizes this mapping by linking each improvement category to the relevant BPR framework components and performance perspectives, providing a synthesized overview of the categories.

Step 3 – Examination of connections among categories: To achieve a generalization for improving business processes based on the analysis discussed above, we hypothesized associations between the 3 misalignment categories (Table 3) and the 10 improvement categories (Table 4), under the assumption that one misalignment category may be solved or reduced by one or more improvement categories. Then, we tested the category associations in IBM SPSS (George and Mallery 2019) using cross-tabulation analysis (Momeni et al. 2018) as shown in Table 5. In fact, for each misalignment situation, a set of one or more improvements was suggested, constituting a total pool of 91 improvements.

After the categorization process, it was observed that a significant portion of the improvements within the same misalignment category were consistently classified together. This indicates a high level of consistency in the classification process and the direction of the relation between the categories.

Table 5 shows the distribution of the selected improvements over the 10 improvement categories across the 3 misalignment categories. This table displays the count and relative percentage of the specific proposed improvements, related to their respective categories, which were proposed for resolving each misalignment category. Italic values indicate the dominant percentages within each improvement category (% within column), highlighting the

Table 3 Misalignment categories

CN	Category type	Six categories of misfit [SV] (Strong and Volkoff 2010) Misfit analytical framework [BG] (van Beijsterveld and van Groenendaal 2016)	Description	Situations
1	Regulation/ procedural issues	Functionality, Role and Organizational culture ^a categories [SV] Process and Perceived misfit type [BG]	Misalignments between the required process and at least one perceived goal (where possibly the process is designed for realizing a different goal)	13
2	Technological/ Implemental issues	Data and Usability categories [SV] Input and output data misfit type [BG]	Misalignments between the IS functionality/usability and the goals it is supposed to support	8
3	KPIs/ Performance assessment issues	Control category [SV] System environment misfit type [BG]	Misalignment between KPIs and process, technology, or other goals	12

^a Organizational culture may influence how rules are enacted, sometimes leading to perception of excessive rigidity, that in turn contributes to perceived goal misalignments

misalignment categories primarily addressed by that improvement category, while bold values indicate the corresponding percentage share of these dominant improvements out of the overall set of proposed improvements (% of total). As an illustration, consider improvement category 9 (Process SLA), which contains 10 specific improvements, five of them (50% – % within column) suggested for addressing misalignment situations classified as misalignment category 1 (Regulation/Procedural issues). Additionally, these 5 improvements account for 5.5% (% of total) of the entire pool of proposed improvements, which amounts to 91 in total, while in total, category 9 (Process SLA) encompasses 11% of the overall count of proposed improvements (10 out of 91). To determine the association between improvement categories and misalignment categories in a way that covers as many categories as possible, we wanted an improvement category to be recommended based on a clear majority (at least 75%) of the misalignment cases it addressed. For most improvement categories this resulted in a mapping to one misalignment category. However, for Category 9, to cover more than 75% it had to be recommended for two misalignment categories (one with 50% of the cases and the next with 30%).

4.2.1 Proposed Framework

Based on the above analysis, we propose the improvement decision framework as shown in Table 6, relating the three misalignment categories and the ten improvement categories.

4.3 Framework Validation and Evaluation

The evaluation of the proposed framework (Table 6) involved a two-step process. Initially, we evaluated the framework through structured interviews with seven BPI experts with scientific and industrial experience, a group size that aligns with recommended ranges for expert elicitation studies and is considered sufficient to ensure diversity while avoiding redundancy of opinions (Krueger et al. 2012). Brief profiles of these experts are provided in Online Appendix 5, Table A. This evaluation utilized both qualitative and quantitative analysis techniques, with the aim of validating the framework. In the second step, we conducted semi-structured interviews with these experts to gauge the framework's utility in terms of improving processes.

Step 1 – Structured interviews with experts: The main goal of this step was to validate the categories of the framework and the associations between them. The evaluation process was focused on testing the extent to which process experts can apply the framework to a set of given situations and reach consistent results. To accomplish this, we used structured interviews with seven BPI experts (Online Appendix 5, Table A). The interviews included open-text parts, which were analyzed qualitatively, and a closed-ended selection part, where the experts matched improvement categories to misalignment categories for a set of 13 scenarios, effectively representing the 3 distinct misalignment categories from each of the case studies. This part, for the seven experts, resulted in 91 matches, which were analyzed quantitatively. Additionally, the interview included the workarounds associated with each of these

Table 4 Improvement categories

CN	Category type	BPR framework components [MR] (Mansar and Reijers 2007) Performance perspectives [VS] (Van Looy and Shafagatova 2016) BPR dimensions [G] (Gross et al. 2021)	Description and additional related references
1	Business KPIs	<i>Internal process performance [VS]</i> : General/Time-related/Cost-related/Flexibility-related/Internal quality <i>Financial performance [VS]</i> : Financial performance for shareholders and top management	Introduction or adjustment of business process KPIs or business metrics for the process and its participants (Dumas et al. 2018; van der Aalst et al. 2016)
2	Process gateways	<i>Behavioral view component [MR]</i> : Knock-out, Parallelism	Policy adjustment and changes in process control gateways and their transition conditions (Satyal et al. 2019; Kahloun and Ghannouchi 2018)
3	Process flow	<i>Behavioral view component [MR]</i> : Resequencing, Exception	Changes in process flow (zur Muehlen and Recker 2013; Gruhn and Laue 2009)
4	Process flexibility	<i>Organization structure component [MR]</i> : Order and Flexible assignment <i>Organization population component [MR]</i> : Empower	Increasing the flexibility of the process and its activities, e.g., decreasing unnecessary complexity and constraints and removing unnecessary mandatory fields and preconditions (Cognini et al. 2018)
5	Responsibility boundaries and specialists	<i>Organization structure component [MR]</i> : Split responsibilities, Customer teams <i>Organization population component [MR]</i> : Specialist, Generalist	Adjustments of activity assignment to the required specialty according to responsibility boundaries of process participants (Kettler et al. 2019; vom Brocke et al. 2014)
6	Split or merge tasks	<i>Operation view component [MR]</i> : Triage, Task composition	Dividing large or general tasks into workable smaller tasks or combining small tasks into one composite or general task (Kahloun and Ghannouchi 2018; Kettler et al. 2019)
7	IS features and capabilities	<i>Technology component [MR]</i> : Integral technology <i>Information component [MR]</i> : Control addition	Implementing or customizing IS features, support tools, and a higher level of documentation as a part of the process (Abdellatif et al. 2018; Ko et al. 2009)
8	Process/IS permission	<i>Organization dimension [G]</i> : Internal participants	Implementation or adjustment of permissions in the process forms and its control flow according to the role of process participants (Cognini et al. 2018; Sadiq et al. 2007)
9	Process SLA	<i>learning and growth Performance [VS]</i> : Employee performance	Adjustments or implementation of service levels and performance times to process activities (Unger et al. 2008; Wetzstein. 2008; van der Aalst et al. 2016; Dasig 2017)
10	Process monitors and alerts	<i>Technology dimension [G]</i> : Infrastructure	Improvements and changes in the level of monitors, indicators, and alerts in the process (Han et al. 2010; Calabro and Lonetti 2015; Chang 2016; Vera-Baquero et al. 2016)

misalignments. Each situation was presented to the experts according to a template that included a situation description, reported workaround motivation, organization and local-unit goals, and workaround consequences. The situation description was followed by two open-ended questions and two closed-ended questions (Online Appendix 5, Table B).

The inclusion of two open-ended questions provided an opportunity for the experts to express their perspectives on any potential goal misalignments within the presented situations. Moreover, these questions facilitated the suggestion of process improvements in response to such misalignments. The two closed-ended questions led the

experts to select the workaround motivation from the 3 misalignment categories and select one or more from the 10 improvement categories. During the interview, we allowed the experts to add new categories at their discretion and as needed. To ensure the experts gained a full understanding of the situations before giving their feedback, each situation was presented and explained to them as a part of the interview. The interviews lasted 90–120 min each, and were conducted using Qualtrics tools (Weber 2021) in online meetings, including recording and transcriptions. The use of this tool facilitated flexibility in conducting the interviews, while also offering efficient and precise data export capabilities to support further

Table 5 A cross-tabulation connecting misalignments to improvements

Misalignment		Improvement										
		1	2	3	4	5	6	7	8	9	10	Total
1	Count	0	10	11	6	10	8	0	1	5	1	52
	% within column	0.0%	83.3%	84.6%	75.0%	100%	100%	0.0%	20.0%	50.0%	11.1%	
	% of total	0.0%	11.0%	12.1%	6.6%	11.0%	8.8%	0.0%	1.1%	5.5%	1.1%	57.1%
2	Count	0	2	2	1	0	0	7	4	2	1	19
	% within column	0.0%	16.7%	15.4%	12.5%	0.0%	0.0%	100%	80.0%	20.0%	11.1%	
	% of total	0.0%	2.2%	2.2%	1.1%	0.0%	0.0%	7.7%	4.4%	2.2%	1.1%	20.9%
3	Count	9	0	0	1	0	0	0	0	3	7	20
	% within column	100%	0.0%	0.0%	12.5%	0.0%	0.0%	0.0%	0.0%	30.0%	77.8%	
	% of total	9.9%	0.0%	0.0%	1.1%	0.0%	0.0%	0.0%	0.0%	3.3%	7.7%	22.0%
T	Count	9	12	13	8	10	8	7	5	10	9	91
	% within column	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
	% of total	9.9%	13.2%	14.3%	8.8%	11.0%	8.8%	7.7%	5.5%	11.0%	9.9%	100%

Table 6 Proposed improvement decision framework

Misalignment category		Improvement category		% within improvement	% within misalignment
1	Regulation/Procedural issues	2	Process gateways	83.3	19.2
		3	Process flow	84.6	21
		4	Process flexibility	75	11.5
		5	Responsibility boundaries and specialist	100	19.2
		6	Split or merge tasks	100	15.3
		9	Process SLA	50	10
2	Technological/Implemental issues	7	IS features and capabilities	100	37
		8	Process / IS permission	80	21
3	KPIs/Performance assessment issues	1	Business KPI	100	45
		9	Process SLA	30	15
		10	Process monitors and alerts	77.8	35

analysis. Validation and clarification of the findings with the domain experts took place as needed. Afterward, we arranged the findings in a two-dimensional matrix, one for the situations and one for the expert feedback. Finally, data analysis and interpretation of the transcribed interviews and statistical analysis of the interview outputs were conducted.

Qualitative analysis: To achieve a comprehensive understanding of the perspective of the experts regarding the misalignment situations, we utilized the insights derived from their responses to the two open-ended questions. This process entailed a qualitative analysis of the perceptions of the experts. We juxtaposed the text presented in the spreadsheet matrix to uncover shared patterns regarding each situation. Our objective was to ascertain common trends and assess the consistency in the

classification of misalignments. The analysis relied on the comparison and identification of words or phrases commonly used by the experts to describe the workaround situations. This included terms such as “conflict,” “misalignment,” and “goal promotion,” as well as words closely related to or associated with the categories we proposed, that is, the reasons behind the workaround and the actions they suggested for each situation to improve the process.

Quantitative analysis: To analyze statistically the findings from the two closed-ended questions (the misalignment and improvement categories), we applied two statistical tests using IBM SPSS (George and Mallery 2019): Fleiss’ kappa, to measure the inter-rater reliability and the agreement level of the experts about the selected categories in each situation, and the Pearson Chi-Square

test of independence with Cross-tabulation analysis, to determine whether there is a significant association between the misalignment and improvement categories.

The hypotheses of the Fleiss' kappa analysis were as follows:

Ha0: There is no significant agreement among the experts about the categories beyond that which would be expected by chance ($\text{kappa} = 0$, due to random chance).

Ha1: There is significant agreement among the experts about the categories that cannot be explained solely by chance ($\text{kappa} \neq 0$, not due to random chance).

The hypotheses of the Pearson Chi-Square test of independence were as follows:

Hb0: There is no association between misalignment and improvement categories.

Hb1: There is an association between misalignment and improvement categories.

Step 2 – Semi-structured interviews with experts: To evaluate the overall usefulness of the framework, we conducted semi-structured interviews with the experts regarding the usefulness and effectiveness of the framework for improving business processes. The interview protocol included three open-ended questions, as shown in Online Appendix 5, Table C.

4.3.1 Evaluation Findings

Step 1 – Structured interviews with experts: The findings of the qualitative analysis indicated a consensus among the majority of experts. They concurred that the primary causes for the workarounds were linked to misalignments among various types of goals. For instance, they articulated the rationale underpinning workarounds using expressions such as “Barriers to goal achievement,” “Goal conflicts,” “Technological constraints or limitations within the process,” etc. Consequently, their recommendations for improvement directions were in general consistent and aligned with this perspective and converged toward our proposed improvement categories. The findings of the quantitative analysis are as follows.

Interpreting the findings of the Fleiss' Kappa analysis: Table 7 shows that the Fleiss' Kappa overall agreement for each category can be considered substantial agreements ($\text{kappa} = 0.727$ and $\text{kappa} = 0.677$) (Masuti and Mohapatra 2015). It appears that the experts substantially agreed on the selection of categories for each presented situation. Furthermore, given that the p -value (0.000) is lower than the chosen significance level ($\alpha = 0.05$), the null hypothesis can be rejected, with a conclusion that there exists a significant agreement, not likely to be due to chance, among the experts ($P_v \leq 0.05$). Additionally, to determine whether there is a category that has a lower agreement level than others, we also analyzed

the Fleiss' kappa agreement level for the individual categories. In this iteration, it is evident that the agreement level for improvement category 6 (split or merge tasks) is lower than for the other categories, and is not statistically significant, having a Kappa value of 0.000 (p -value > 0.05). This suggests that the experts had differing opinions or interpretations in the case of improvement category 6. According to those findings, we returned to the interview recordings and the findings with the aim of determining whether the domain experts found this category to be redundant or referenced it as a part of other categories. In fact, we found that the domain experts included category 6 as a part of category 5 (responsibility boundaries) in their answers, and accordingly those categories were unified to one improvement category, as shown in Table 8.

Interpreting the findings of the Pearson Chi-Square test of independence: Given that the p -value (0.001) is lower than the selected significance level ($\alpha = 0.05$), the null hypothesis can be rejected. This indicates a significant association between misalignment and improvement categories. That is, it suggests that a particular misalignment category can be effectively addressed by one or more improvement categories in a significant manner. For detailed understanding of which specific misalignment categories can be solved by which specific improvement categories, we conducted a Cross-tabulation analysis, as shown in Table 9.

As can be seen in Table 9, each improvement category can mainly be used for specific misalignment categories, and for each misalignment category the higher-frequency improvement categories form possible improvement directions (italicized in the table), all with statistical significance. To summarize the findings, the validated categories, along with their respective Fleiss' kappa values and identified associations among them, allow us to validate the improvement decision framework as presented in Table 10 below.

Step 2 – Semi-structured interviews with experts: The experts did not endorse a single standardized method for process improvement. They acknowledged that the methods employed often vary, contingent on the type of organization and the complexity of the process. This applies from the initial stages of learning and analyzing the process to uncovering improvement opportunities, using a combination of qualitative analysis techniques, such as interviews and observations, and quantitative analysis methods, such as PM and queries to assess compliance with process KPIs, execution times, bottlenecks, and other issues. These methods continue to be employed throughout the improvement stage, where improvement patterns may be considered as needed. After the experts used the framework for process improvement, they explained that the framework was very useful and effective in process analysis and

Table 7 Statistical analysis

Test	Purpose	Findings	Interpretation
Fleiss' kappa	Overall agreement on Misalignment categories	$k = 0.727, P < 0.000$ ($P_v \leq 0.05$)	Substantial (strong) agreements
	Overall agreement on Improvement categories	$k = 0.677, P < 0.000$ ($P_v \leq 0.05$)	Substantial (strong) agreements
Pearson Chi-Square	Association between misalignment and improvement categories	$\chi^2(16,265) = 301.960$ $P < 0.001$ ($P_v \leq 0.05$)	One misalignment category can be solved by one or more improvement categories

Table 8 The unified category

CN	Category type	Description
5 + 6	Process operation	Dividing large or general tasks into workable smaller tasks or combining small tasks into one composite or general task, respectively to the required specialty and to responsibility boundaries of process participants (Mansar and Reijers 2007; Kahloun and Ghannouchi 2018; Kettler et al. 2019)

Table 9 Distribution of misalignments over improvements

Misalignment		Improvement									
		1	2	3	4	5 + 6	7	8	9	10	Total
1	Count	4	21	34	20	36	4	7	15	5	146
	% within column	12.1%	95.5%	97.1%	90.9%	87.8%	8.9%	29.2%	71.4%	22.7%	55.1%
2	Count	1	1	0	2	0	41	15	2	5	67
	% within column	3.0%	4.5%	0.0%	9.1%	0.0%	91.1%	62.5%	9.5%	22.7%	25.3%
3	Count	28	0	1	0	5	0	2	4	12	52
	% within column	84.8%	0.0%	2.9%	0.0%	12.2%	0.0%	8.3%	19.0%	54.5%	19.6%
T	Count	33	22	35	22	41	45	24	21	22	265
	% within column	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Table 10 The improvement decision framework

Misalignment category		Improvement category	
1	Regulation/Procedural issues ($k = 0.786, P < 0.000$)	2	Process gateways ($k = 0.364, P < 0.001$)
		3	Process flow ($k = 0.728, P < 0.000$)
		4	Process flexibility ($k = 0.497, P < 0.000$)
		5 + 6	Process operation ($k = 0.739, P < 0.000$)
		9	Process SLA ($k = 0.535, P < 0.000$)
2	Technological/Implemental issues ($k = 0.772, P < 0.000$)	7	IS features and capabilities ($k = 0.671, P < 0.000$)
		8	Process/IS permission ($k = 0.645, P < 0.000$)
3	KPIs/Performance assessment issues ($k = 0.604, P < 0.000$)	1	Business KPI ($k = 0.829, P < 0.000$)
		10	Process monitors and alerts ($k = 0.825, P < 0.000$)

improvement, providing an immediate focus on problems that need to be solved for improving the process. In

addition, the framework gave them a comprehensive understanding of the workaround situations in a relatively

short time, which finally directed them to the improvements themselves. An example from the text, “I found the framework very practical, useful and instructive regarding what to focus on in order to improve the process ...” and “In my opinion, this framework is so useful and intuitive in identifying the various misalignment situations, their workarounds, and finally improving the process accordingly...” To enhance the output of the framework, the experts recommended adapting the list of improvements to the specific organizational context by involving relevant stakeholders or key users who possess an understanding of the organizational environment and dynamics. An example from the text, “In the proposed solutions, I will propose considering also the organizational politics and atmosphere that may vary from one organization to another” and “I would suggest involving a user from within the organization who will examine the improvements in the organizational environment.” In practice, the experts noted that the framework guided both the identification of misalignments and the selection of relevant improvement categories. For example, one expert said: “the framework showed me which improvement to focus on in order to improve the process, but shaping the solution still depended on my understanding of the organization.” They used their professional experience to refine and tailor the resulting improvement actions to the organizational context.

In summary, the two steps complemented each other, as experts initially validated the categories and their associations, and subsequently evaluated the overall effectiveness of the framework.

5 The Overall Improvement Cycle

To leverage knowledge about workarounds and their causes for process improvements, we propose a streamlined end-to-end process improvement cycle, ranging from detecting and analyzing workaround and misalignment situations to the proposal of tailored process improvements to organizations. This cycle, including its steps and tools, was developed and iteratively refined through the case studies reported above, to support a complete and systematic procedure for process improvement. We begin by illustrating the proposed workaround-driven improvement cycle (WDIC) and its associated components. We then demonstrate and validate the WDIC using a fourth case study and subsequently evaluate the usefulness of its results in real organizational contexts.

5.1 Workaround-driven Improvement Cycle (WDIC)

The proposed cycle, illustrated in Fig. 6, has four main steps: data collection, workaround detection, misalignment

analysis, and process improvement, as discussed below. These four steps can be repeated periodically, thus forming a cycle of ongoing process improvement.

Step 1 – Data collection: This step includes collection and preparation of data required to initiate process improvement. Following the description of the case studies, the collected data should include process and IS documentation and event logs, as well as data collected using qualitative techniques, such as interviews and observations. Next, an initial analysis of the gathered data is essential for preparing and creating the main tools for subsequent steps. This involves the creation of as-is process models and i* models to represent the actors, goals, and dependencies in relation to the process tasks.

Step 2 – Workaround detection: This step starts with an initial set of workarounds, identified through interviews and observations (step 1). This list is extended and triangulated in two ways. First, the goal-based analysis (Outmazgin et al. 2020) identifies conflicts and misalignments in the i* models. These may relate to already known workarounds or indicate additional unreported ones. Second, process mining techniques can serve for detecting workarounds based on the actual behavior reflected in event log data. We note that while the three case studies used only basic process discovery, case study 4 employed the more sophisticated SWORD (van der Waal et al. 2025) framework for this purpose, as reported in Outmazgin et al. (2023). These two approaches are complementary, as neither is capable of independently identifying all workarounds. In this step, we specifically focus on identifying recurrent workarounds as patterns that appear repeatedly under similar circumstances, as these reflect persistent misalignments and form the core of our analysis. The resulting set of identified workaround patterns provides an understanding of how they are performed and by which actors, offering additional valuable insights that can be used to develop targeted process improvement actions.

Step 3 – Root cause analysis: Considering the detected workarounds, this step uses the TPB-based motivational model (Soffer et al. 2023) for analysis and understanding of the root causes (e.g., goals misalignment) that motivate process participants to perform the detected workaround as a means to achieve their goals. Based on the findings of this analysis, each misalignment situation will be classified into one of the three predefined misalignment categories, as outlined in Table 3. The outcome of this step is a categorized list of misalignment situations, each mapped to its corresponding category.

Step 4 – Improvement selection and implementation: This step focuses on utilizing the improvement decision framework (Table 10) to select appropriate improvement categories for each classified misalignment situation. The structured framework ensures that every misalignment

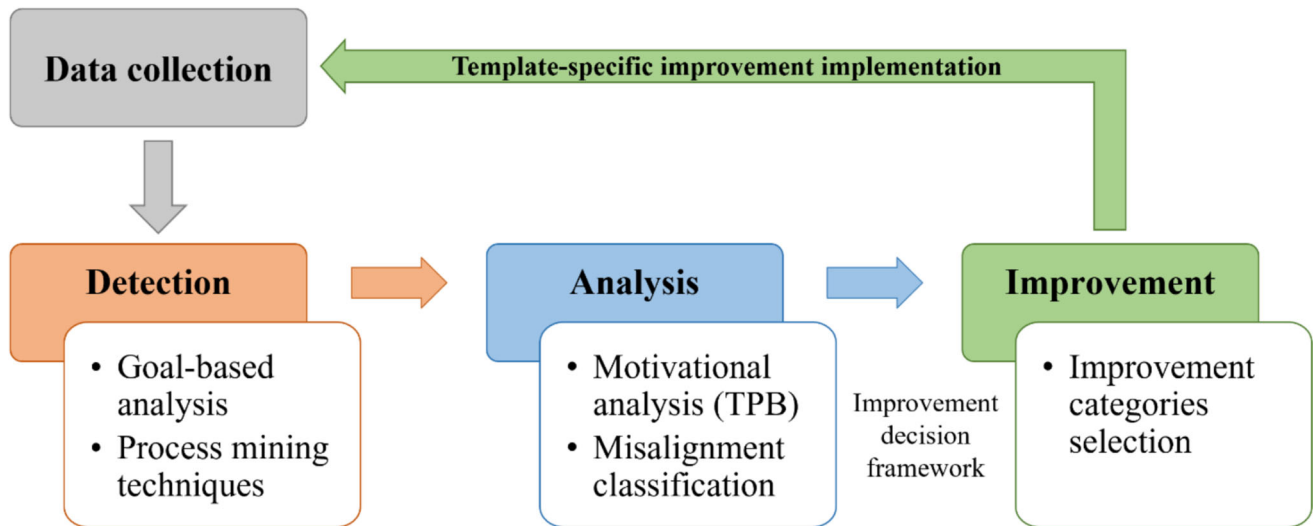


Fig. 6 Workaround-driven improvement cycle (WDIC)

situation is systematically linked to the most relevant improvement strategy, thereby facilitating the design of effective, context-specific solutions based on the insights gained during the analysis. This step is fully based on the validated associations between the misalignment and improvement categories. To translate the selected improvement categories into concrete and tailored process improvements that can be implemented, we propose the use of a structured template (see Table 11) that captures all necessary information required to develop specific

improvement actions based on the workaround-based improvement framework. This template supports the transformation of abstract improvement categories into actionable solutions by leveraging key findings from the workaround analysis. Along with these proposed improvements, it is possible to also consider the existing workaround as a permanent solution. However, for the workaround to be an acceptable solution, it should be carefully considered whether, while resolving a locally perceived misalignment, it does not hamper any other goal

Table 11 Template: specific improvement implementation (an example)

Workaround detected	Sales negotiation activities performed manually and not in the system (Online Appendix 4, Table A, Situation 1.2 and 1.3)
Workaround description	The customer manager prepares and sends price proposals to customers manually before they exist in the IS. The customer manager uses Word documents or Excel spreadsheets and sends them to the customers directly in regular email, and thus, these actions are not documented in the IS
Underlying misalignment	Misalignment between the customer manager's goals to respond quickly and flexibly and to satisfy the customer, as expected in this role, and the goal of management to avoid sending unapproved proposals, with the aim of reducing organizational risks. If the customer manager waited for approval for each new price proposal, as well as for each change introduced after the approval, the goal of quick and flexible response would not be achieved, and customer satisfaction may be reduced
Result	Harm can be caused to the organization by sending unapproved price proposals that might create a commitment to unprofitable or risky deals
Proposed solution	We proposed improvements to reduce the strictness of the procedure of approvals and approval rounds. Because the misalignment in this situation was related to procedural aspects, we based the solution on improvement categories of process gateways, process flow, and process flexibility, as follows <i>Process gateways</i> – introducing threshold conditions for the amount and the profitability rate of the price proposals that require approval, which would increase the degree of freedom for the customer managers and enable quick processing of small price proposals <i>Process flow</i> – introducing several different approval trails that can be selected according to price proposal amounts and the profitability rate and considering whether a new price proposal is involved or a change in an existing one <i>Process flexibility</i> – changing the procedure to allow the customer manager to send an unapproved price proposal to the customers with a warning stating that the price proposal commits the organization only after full approval

of the organization. The use of the template, as well as the considerations regarding existing workarounds, are exemplified in Table 11. The example is based on one of the recurrent workaround patterns identified in Case Study 1, using the qualitative descriptions and classifications documented in Online Appendix 4, Table A.

The example in Table 11 demonstrates how a workaround revealed a fundamental misalignment between the need for prompt and flexible customer engagement and the organization's emphasis on risk mitigation through formal approval mechanisms. Rather than adopting the workaround as-is, which could institutionalize non-compliant behavior and weaken organizational controls (Alter 2014; Debono et al. 2013), the process was redesigned based on the insight it revealed. The proposed improvements included the introduction of threshold-based approval conditions and multiple approval pathways, allowing greater process flexibility without compromising managerial oversight. This case exemplifies how a workaround can inspire a solution, as it may be adopted as an official practice in conditions when the risk is not high, namely below a given threshold, but not otherwise.

5.2 Demonstration and Validation

For demonstrating and validating the proposed WDIC we conducted an additional, independent, case study (Case study 4). This enabled us to evaluate and confirm the general applicability and effectiveness of the WDIC. This case study was conducted within a separate division of Organization 3, as detailed in Table 1 above. The selected process, the *Purchase Order* process, serves as a core process central to all purchasing activities. It is systematically designed with structured process flows that incorporate controls and transition rules based on predefined monetary thresholds and approval cycles. This key process supports five departments and involves dozens of participants across its various stages. It is considered critical for achieving both economic and operational efficiency, as well as ensuring compliance with regulatory requirements expected from a public organization. The process is managed using an industry-standard ERP system entirely, separate from the one previously studied, involving different stakeholders and participants. An overview of the case study is provided below.

All four steps of the WDIC were conducted as follows: (1) Data collection – included collection of IS structural and procedural documentation, IS screenshots, existing process models, outputs of interviews and observations, and an event log, comprising 38333 events (5908 cases) covering a three-year period (Online Appendix 1, Table B). This was analyzed to comprehensively understand the process and create both BPMN and i* models (Online

Appendix 2 and 3). Interviews were conducted with process participants and stakeholders following the interview guide. (2) Workaround detection – Workarounds reported in the interviews were triangulated using a goal-based analysis approach, complemented by process mining (PM) techniques. In total, 9 workaround patterns were identified in this process. (3) Root cause analysis – Each detected workaround was analyzed using the motivational model (TPB) and the i* models to uncover their underlying root causes. Consequently, 11 misalignments were discovered and categorized into the three misalignment categories. (4) Improvement proposal – Finally, targeted improvements were proposed for each identified misalignment situation based on recommendations provided by the improvement decision framework (improvement categories). These improvements were subsequently structured and suggested for implementation using the template presented in Table 11. The identified misalignments and selected improvement directions are summarized in Table 12.

After discussions in the organization, assessing the potential benefit and cost of the proposed improvements, 14 improvements were implemented to address 6 of 9 workarounds, which were identified by the stakeholders as having higher importance and occurred in approximately 65% of the process instances. Four of these workaround patterns are presented in detail in Table 13, illustrating their operational manifestation, recurrence, underlying misalignments, and the corresponding improvement directions derived through the WDIC.

The improvements were further assessed through focused interviews and follow-up meetings with process participants and relevant stakeholders after a period of one year. According to these stakeholders, the process demonstrated clear improvements in both efficiency and goal alignment, significantly reduced workaround occurrences and enhanced compliance with formal procedures. We note that this assessment reflects stakeholder perception of the WDIC results without a full comparative measurement. In summary, stakeholder feedback reinforced the practical utility and effectiveness of the WDIC, confirming its capability to deliver substantial and sustainable business process improvements in real-world organizational contexts.

5.3 WDIC Evaluation

To evaluate the practical usefulness of the proposed WDIC within an organizational context, we conducted two series of meetings and discussions with relevant stakeholders and managerial staff from case studies 1, 3, and 4 (case study 2 was excluded due to limited availability for evaluation). This was done in two phases. In the first phase the suggested improvements were presented to process

Table 12 Misalignments and improvements matrix in case study 4

Misalignment	Improvement					
	1 Business KPI	2 Process gateways	3 Process flow	7 IS features & capabilities	9 Process SLA	10 Process monitors
1 Procedural		1	2		3	
2 Technological				3		
3 KPIs	5					4

Table 13 Representative workarounds and their corresponding improvement directions

Workaround pattern	Recurrence (Event Log)	Underlying misalignment (Category)	Improvement directions (Framework)
W1: Retrospective purchase initiation Description: Requests for purchase are submitted only after work has already begun, bypassing the formal approval trail	2304 (39%)	C3: KPI and performance assessment misalignment Misalignment between departmental performance goals and procurement process timelines	C1: Align KPIs across units C10: Introduce monitoring dashboards and alerts
W2: Advancing supplier engagement before the approval trail is completed Description: Employees engage with suppliers before the required approvals are completed, effectively advancing the process outside the formal approval trail	3604 (61%)	C1: Regulatory and procedural misalignment C3: KPI and performance assessment misalignment Pressures to meet service expectations despite procedural constraints	C9: Define clear SLA expectations C1: Synchronize approval dependencies (KPIs) C10: Establish transparency mechanisms for tracking approval status
W3: Use of email-based approvals outside the system Description: Approvals and price proposals are exchanged via email rather than recorded within the system. Only after an email approval is obtained is the purchase entered into the system for the formal approval flow	4372 (74%)	C2: Technological and implementational misalignment IS limitations preventing full documentation and processing	C7: Extend system capabilities (uploading, comparison tables, structured messaging), enforce IS-based documentation, and define clear usage rules
W4: Catalog-based approval bypass Description: Items or services are entered into the request using a vendor-approved catalog in order to bypass the system's validation rules and cumulative purchasing constraints. After the request is approved, employees coordinate directly with the supplier to obtain the products that are actually required	946 (16%)	C2: Technological and implementational misalignment System validation rules and cumulative purchasing constraints are misaligned with employees' operational needs ^a	C7: Introduce flexible and transparent system controls, such as exception-request mechanisms, configurable thresholds, catalog governance, and enhanced reconciliation during goods receipt

a Similar patterns may also arise from attempts to bypass procurement rules or tender regulations, where increasing flexibility in the procedure could expose the organization to legal risks

stakeholders and management in frontal meetings, following the template-specific implementation structure detailed in Table 11. The second phase took place after a period of one year, when we revisited the organizations to obtain feedback from the stakeholders and management. This feedback was sought to assess which of the improvements that had been suggested was implemented and still operationally practiced as suggested in a

stable manner over time. The findings are summarized in Table 14.

In summary, both stakeholders and the organizational management of the participating organizations found the solutions proposed through the WDIC to be highly useful. Yet, due to distinct organizational factors, such as resource availability, professional staffing levels, and budgetary constraints, each organization adopted a tailored

Table 14 Improvements implementation findings

WDIC	Case study 1 Sales forecast	Case study 3 Student registration	Case study 4 Purchase order
Workaround patterns	11	14	9
Misalignment situations	13	16	11
Proposed improvements	12	13	18
Implemented improvements after 1 year	8	9	10

implementation plan, prioritizing some of the suggested improvements.

6 Discussion

This paper introduces an innovative approach to improving business processes by detecting and analyzing workarounds, uncovering their underlying reasons, and targeting these reasons in process improvement. This is proposed as a repeating course of action, whose repetition forms an improvement cycle. While workarounds are not the sole source of insight in real-world process improvement, they provide a practical and systematic entry point for uncovering process issues that may remain unnoticed through conventional analysis. As such, the WDIC complements existing BPM practices by guiding continuous detection of underlying misalignments. To establish the contribution of the proposed approach, we discuss the similarity and differences of elements in this approach with existing literature.

Previous research streams have explored the motivations for actions deviating from organizational policy to achieve legitimate goals at various levels, developing and adopting explanations and theories (e.g., Alter 2014; Haag et al. 2022). We rely on the TPB-based motivational model (Soffer et al. 2023) as a systematic and theory-based way of navigating such conflicts. This model is more comprehensive than other theories used in this context, encompassing technical, societal and individual motivational elements as well as perceptions of control. It hence provides a basis that allows explaining the full extent of nuances of the data we collected. In addition to its role as a theoretical underpinning, this model also guides an additional, detailed analysis that complements the goal-based analysis (Outmazgin et al. 2020), used primarily for detecting and analyzing workarounds and misalignments. Building on the idea that workarounds are motivated by perceived misalignments and misfits is also not new, and was suggested by, e.g., Alter (2015a, b), Strong and Volkoff (2010), Outmazgin et al. (2020), and others. Our approach takes this observation a step forward and, using the improvement decision framework, supports a smooth

transition from identified misalignments to process improvement directions.

Previous studies have already attempted to use work-around knowledge to improve business processes [e.g., Röder et al. (2015), van de Weerd et al. (2019) and Beerepoot et al. (2019a)]. However, the studies were focused on investigating workarounds and assessing them so this knowledge could be used as input for process improvement design and decisions. In fact, the solutions that could be derived would concentrate primarily on observed workarounds which could be adopted as official practices, without delving into the underlying motivations driving the process participants to perform them. Our approach, in contrast, treats the observed workarounds as a symptom of a deeper problem to be solved. While actual workarounds may be considered as possible solution directions, they are not necessarily the best ones. Rather, our focus is on solving the problem revealed through workaround analysis, and the improvement decision framework provides a structured way for developing solutions.

In a different vein, our approach also builds on studies that focused on developing process improvement directions detached from workarounds [e.g., Mansar and Reijers (2007), Van Looy and Shafagatova (2016), and Gross et al. (2021)]. These studies were focused on improvement directions as solutions and did not specifically address the comprehensive investigation that is needed to pinpoint the problems that would lead to exact required improvements. Rather, they often provided sets of guidelines or best practices for implementing recommended improvement patterns in certain situations. Nevertheless, to use these guidelines, a process investigation is required. This paper serves as a bridge between these two research streams by providing a holistic and systematic approach for improving business processes, encompassing both problems that are reflected workarounds and solutions.

In summary, while various elements of the WDIC have been addressed in the literature, a main contribution is proposing a streamlined end-to-end approach, starting with the identification and investigation of specific areas in the process that require improvement (reflected by workarounds) and concluding with the proposal of structured

and tailored improvements. As mentioned, our approach concentrates on workaround patterns, namely, deviations that consistently reappear across similar situations and therefore signal structural misalignments. This focus ensures that the improvement cycle addresses persistent issues rather than incidental deviations. By understanding the root causes of these workarounds, the approach can contribute to the development of targeted process improvements that address them, while incorporating the insights gained from workaround investigation, thereby reducing the need for workarounds. We acknowledge that new workarounds may emerge over time due to organizational changes, technological evolution, or external events. This reflects the dynamic nature of business processes and the phenomenon of process drift (Yeshchenko et al. 2019). Accordingly, our WDIC is designed to be applied iteratively, allowing organizations to monitor and address emerging misalignments as part of a continuous improvement cycle. In practical terms, this iterative cycle complements existing BPM lifecycle practices (Dumas et al. 2018), as detecting a workaround during the monitoring phase would trigger a new cycle in which workaround root-cause analysis initiates the design of targeted process improvements. To further demonstrate the applicability of the WDIC, we included several representative recurrent workarounds in Case Study 4 together with the misalignments underlying them and the improvement actions derived. These concise examples illustrate how the WDIC operationalizes the transition from workaround detection to actionable process improvements.

While the WDIC guides a systematic course of process improvement, its application in practice may benefit from the following lessons we learned while conducting the case studies reported in the paper:

1. Workarounds can be identified through various channels and techniques, including qualitative methods such as interviews and observations, as well as quantitative methods such as PM. However, the underlying causes that motivate these workarounds are not always easily identified. Focusing on misalignments provides an appropriate and rather easy way to achieve this.
2. In almost every instance of revealed misalignment, one or more workarounds can be discovered, and we have not observed any workarounds that did not have an associated misalignment. In other words, workaround and misalignment analysis complement each other.
3. The investigation of misalignments and their classification are crucial steps in the framework. Incorrect classification may lead to process improvements that would not effectively address the real problems.

4. To examine the effectiveness of the improvements proposed using the improvement decision framework and to achieve an ongoing process improvement, it is recommended to reinitiate the detection and analysis of workarounds periodically.

We note that the diversity of case studies used in this paper played a significant role in the development of a versatile approach. Therefore, the WDIC can be applied in various organizations to improve their business processes, transcending limitations associated with specific domains or industries. As the WDIC examines goals at various levels as part of process improvement, it can enable organizations to align their strategic and operational goals with the process tasks intended to accomplish them. Consequently, this can lead to a reduction in workarounds and increase organizational efficiency and effectiveness using the process.

6.1 Limitations

This research has several limitations that should be acknowledged concerning the study and the proposed approach.

First, because of the qualitative nature of the case studies and the limited number of organizations involved, a question of transferability as well as of the completeness of the framework arises. We attempted to alleviate this concern by conducting an overall evaluation of the decision framework involving BPI experts (although in a limited number), by complementing the empirical findings with relevant literature, and by validating the entire WDIC in an independent case study. Nevertheless, it is possible that it will be extended in future based on additional studies.

Second, the processes selected for the case studies were those receiving high managerial attention. While this could potentially lead to higher process adherence due to closer oversight, it might also encourage workaround behavior when employees prioritize performance targets or managerial expectations over formal procedures. Thus, the effect of selecting such processes on workaround behavior is ambiguous and should be considered a contextual limitation of this study.

Third, the applicability of the proposed approach is impaired due to the goal-based analysis approach, which requires knowledge in creating the i^* models. As explained, these models have an important role in the analysis of misalignment situations. Consequently, the analysis results can significantly affect the overall effectiveness of the approach in proposing improvements. Although guidelines have been proposed on how to create the i^* models (Outmazgin et al. 2020), different models may still be created by different analysts. Furthermore, the

model employs two qualitative levels of negative contributions (*Hurt or Break*) without differentiating between them. It seems that additional semantic guidelines can be developed for using the contribution level in a way that will support prioritization. Alternative goal models can also be used if they are found effective in analyzing goal misalignments.

Finally, the proposed approach is focused solely on improvements related to workarounds. Hence, weaknesses or problems in the process that are not manifested as workarounds will not be addressed by this approach. We acknowledge that other types of process deviations, such as fraud or deliberate violations, may also reflect existing misalignments (Röder et al. 2016). However, these are driven by fundamentally different intentions and require different detection and mitigation mechanisms (Murphy and Free 2016). The workarounds we address are typically intentional but non-malicious deviations, aimed at enabling work under perceived constraints. As such, they are observable, often repeated, and initiated by end users seeking to achieve process goals despite structural or procedural misfits (Beerepoot and van de Weerd 2018). Our approach, therefore, is tailored to leverage the constructive nature of workarounds rather than to address all forms of non-compliance.

7 Conclusions

In recent years, many efforts have been devoted to the development of approaches for improving business processes. In this study, a systematic approach was developed to improve business processes by detecting, analyzing, and learning workarounds and their reasons. The identification of workaround situations within a process can serve as an indicator of improvements needed in the process, while the comprehensive knowledge and root understanding of what drives those workarounds can guide the required process improvements. The proposed approach provides a systematic procedure for process improvement based on a systematic investigation of workaround situations. Based on this investigation and the workaround knowledge, the approach facilitates the selection of appropriate improvement patterns aligned with the revealed reasons (the misalignments). This, in turn, enables targeted design and implementation of the most appropriate process improvements.

The contributions of the paper are both to research and practice. It contributes to two existing bodies of research, workaround research and business process improvement research, and combines them. While the main concrete contribution is the approach that supports a systematic process improvement based on workarounds, the novelty of

this approach also stems from its theoretical basis and depth of associated analysis. Furthermore, as a part of the WLDC, the decision framework ties misalignment categories with process improvement directions and can thus be applied even when no workarounds are known to exist. This framework and the systematic WDIC that leads from observed workarounds to process improvement is also a substantial contribution to practice, as it can be adopted and applied by organizations that seek process improvement. As indicated by the experts we interviewed, it is highly practical and helps focus improvement efforts on specific problems to be solved.

In the future, the proposed approach should be examined in additional research studies, which could involve employing quantitative techniques for detecting workaround and misalignment situations. For instance, the utilization of a quantitative scale to rate the negative contributions (e.g., goals misalignment) could be explored as an alternative to the two qualitative levels of “*Hurt*” and “*Break*.” This research direction could potentially lead to a reduction in the labor-intensive efforts required for the detection and analysis of workaround and misalignment situations. Moreover, quantitative techniques may help in prioritization of the revealed misalignment situations. This prioritization could, in turn, guide the use of the approach to recommend necessary process improvements based on these quantitative assessments.

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